practice questions for AP practical

1. Create an android application to pass the data from one activity to another activity in the same application using intent (P3)
-->
1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows

```
package com.example.quest1

import android.content.Intent
import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import androidx.compose.ui.platform.LocalContext

class MainActivity : ComponentActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    setContent {

      MainScreen()

    }

}
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun MainScreen() {
val context = LocalContext.current
var name by remember { mutableStateOf("") }

Column(
    modifier = Modifier

        .fillMaxSize()

        .padding(16.dp)

){
```

```
    Text(
        text = "Main Activity",
        style = MaterialTheme.typography.headlineLarge
    )
    Spacer(modifier = Modifier.height(20.dp))

    OutlinedTextField(
        value = name,
        onValueChange = {name = it},
        label = {Text("Enter Your Name: ")}
    )

    Spacer(modifier = Modifier.height(20.dp))

    Button(onClick = {
        val intent = Intent(context, SecondActivity::class.java)
        intent.putExtra("username", name)
        context.startActivity(intent)
    }){
        Text("Send Data")
    }
}
}
```

2. Creating SecondActivity.kt (Steps)

Right click java → com.example.activitydemo

Click New → Kotlin Class/File

Name the file SecondActivity

Select Class and click OK

3. Modifying SecondActivity.kt

Edit the SecondActivity.kt file as follows:

```
package com.example.quest1

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp

class SecondActivity : ComponentActivity() {

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)



    val receivedName = intent.getStringExtra("username")



    setContent {

      SecondScreen(receivedName ?: "")

    }

}
}

@Composable
fun SecondScreen(name: String){

Column(
    modifier = Modifier
```

```
        .fillMaxSize()

        .padding(16.dp)

){

    Text(

        text = "Second Activity",

        style = MaterialTheme.typography.headlineLarge

    )



    Spacer(modifier = Modifier.height(20.dp))



    Text(

        text = "Received Data: $name",

        style = MaterialTheme.typography.bodyLarge

    )

}
}
```

4. Modifying AndroidManifest.xml

Add SecondActivity inside the <application> tag.

```
<activity android:name=".SecondActivity"/>
```

5. Configuring build.gradle.kts (Module:app)

```
plugins {
id("com.android.application")
id("org.jetbrains.kotlin.android")
}

android {
namespace = "com.example.quest1"
compileSdk = 35
```

```
defaultConfig {
    applicationId = "com.example.quest1"

    minSdk = 24

    targetSdk = 35

    versionCode = 1

    versionName = "1.0"



    testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"

    vectorDrawables {

        useSupportLibrary = true

    }

}

buildTypes {
    release {

        isMinifyEnabled = false

        proguardFiles(

            getDefaultProguardFile("proguard-android-optimize.txt"),

            "proguard-rules.pro"

        )

    }

}
compileOptions {
    sourceCompatibility = JavaVersion.VERSION\_1\_8

    targetCompatibility = JavaVersion.VERSION\_1\_8

}
kotlinOptions {
    jvmTarget = "1.8"
```

```
}
buildFeatures {
    compose = true

}
composeOptions {
    kotlinCompilerExtensionVersion = "1.4.3"

}
packaging {
    resources {

        excludes += "/META-INF/{AL2.0,LGPL2.1}"

    }

}
}

dependencies {

implementation("androidx.core:core-ktx:1.9.0")
implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
implementation("androidx.activity:activity-compose:1.10.0")
implementation(platform("androidx.compose:compose-bom:2023.03.00"))
implementation("androidx.compose.ui:ui")
implementation("androidx.compose.ui:ui-graphics")
implementation("androidx.compose.ui:ui-tooling-preview")
implementation("androidx.compose.material3:material3")
implementation("androidx.appcompat:appcompat:1.6.1")
implementation("com.google.android.material:material:1.8.0")
implementation("androidx.constraintlayout:constraintlayout:2.1.4")
testImplementation("junit:junit:4.13.2")
androidTestImplementation("androidx.test.ext:junit:1.3.0")
androidTestImplementation("androidx.test.espresso:espresso-core:3.7.0")
androidTestImplementation(platform("androidx.compose:compose-bom:2023.03.00"))
androidTestImplementation("androidx.compose.ui:ui-test-junit4")
debugImplementation("androidx.compose.ui:ui-tooling")
debugImplementation("androidx.compose.ui:ui-test-manifest")
}
```

7. Configuring gradle-wrapper.properties

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

2. Create a user interface with a TextView, Button, and ImageView. When the button is clicked, the image should be changed dynamically
-->
1. Modifying MainActivity.kt

```
package com.example.quest2

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.Image
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.res.painterResource
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    setContent {

        MainScreen()

    }

}
}

@Composable
fun MainScreen() {

var imageChanged by remember { mutableStateOf(false) }

Column(
    modifier = Modifier

        .fillMaxSize()

        .padding(16.dp),

    horizontalAlignment = Alignment.CenterHorizontally

){
```

```
    Text(
        text = "Dynamic Image Change",
        style = MaterialTheme.typography.headlineLarge
    )

    Spacer(modifier = Modifier.height(20.dp))

    Image(
        painter = painterResource(
            id = if(imageChanged) R.drawable.wp2 else R.drawable.wp
        ),
        contentDescription = "Dynamic Image",
        modifier = Modifier.size(200.dp)
    )

    Spacer(modifier = Modifier.height(20.dp))

    Button(onClick = {
        imageChanged = !imageChanged
    }){
        Text("Change Image")
    }
}
}
```

2. Adding Images to Drawable (Steps)

Open res → drawable folder.

Add two images.

Example:
image1.png
image2.png

These images will be used to dynamically change the ImageView when the button is clicked.

3. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
id("com.android.application")
id("org.jetbrains.kotlin.android")
}

android {
namespace = "com.example.quest2"
compileSdk = 35

defaultConfig {
    applicationId = "com.example.quest2"

    minSdk = 24

    targetSdk = 35

    versionCode = 1

    versionName = "1.0"



    testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"

    vectorDrawables {

      useSupportLibrary = true

    }

}
```

```
buildTypes {
    release {

        isMinifyEnabled = false

        proguardFiles(

            getDefaultProguardFile("proguard-android-optimize.txt"),

            "proguard-rules.pro"

        )

    }

}
compileOptions {
    sourceCompatibility = JavaVersion.VERSION\_1\_8

    targetCompatibility = JavaVersion.VERSION\_1\_8

}
kotlinOptions {
    jvmTarget = "1.8"

}
buildFeatures {
    compose = true

}
composeOptions {
    kotlinCompilerExtensionVersion = "1.4.3"

}
packaging {
    resources {

        excludes += "/META-INF/{AL2.0,LGPL2.1}"

    }

}
}

dependencies {

implementation("androidx.core:core-ktx:1.9.0")
implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
```

```
implementation("androidx.activity:activity-compose:1.10.0")
implementation(platform("androidx.compose:compose-bom:2023.03.00"))
implementation("androidx.compose.ui:ui")
implementation("androidx.compose.ui:ui-graphics")
implementation("androidx.compose.ui:ui-tooling-preview")
implementation("androidx.compose.material3:material3")
implementation("androidx.appcompat:appcompat:1.6.1")
testImplementation("junit:junit:4.13.2")
androidTestImplementation("androidx.test.ext:junit:1.3.0")
androidTestImplementation("androidx.test.espresso:espresso-core:3.7.0")
androidTestImplementation(platform("androidx.compose:compose-bom:2023.03.00"))
androidTestImplementation("androidx.compose.ui:ui-test-junit4")
debugImplementation("androidx.compose.ui:ui-tooling")
debugImplementation("androidx.compose.ui:ui-test-manifest")
}
```

4. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

3. Implement a custom color scheme for your registration form. Set the background color of the entire layout to a light color, and use a contrasting color for the Submit button.

-->

1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

package com.example.quest3

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.*
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.unit.dp
```

```
class MainActivity : ComponentActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    setContent {

        RegistrationForm()

    }

}
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun RegistrationForm() {
Column(
    modifier = Modifier

        .fillMaxSize()

        .background(Color.Yellow)

        .padding(16.dp),

    horizontalAlignment = Alignment.CenterHorizontally

){
    Text(

        text = "Registration Form",

        style = MaterialTheme.typography.headlineLarge

    )



    Spacer(modifier = Modifier.height(20.dp))



    OutlinedTextField(

        value = "",

        onValueChange = {},
```

```
        label = {Text("Enter Name")}
    )

    Spacer(modifier = Modifier.height(10.dp))

    OutlinedTextField(
        value = "",
        onValueChange = {},
        label = {Text("Enter Email")}
    )

    Spacer(modifier = Modifier.height(20.dp))



    Button(
        onClick = {},
        colors = ButtonDefaults.buttonColors(
            containerColor = Color.Magenta
        )
    ){
        Text("Submit", color = Color.White)
    }
}
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
id("com.android.application")
id("org.jetbrains.kotlin.android")
}

android {
namespace = "com.example.quest3"
compileSdk = 35

defaultConfig {
    applicationId = "com.example.quest3"

    minSdk = 24

    targetSdk = 35

    versionCode = 1

    versionName = "1.0"



    testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"

    vectorDrawables {

        useSupportLibrary = true

    }

}

buildTypes {
    release {

        isMinifyEnabled = false

        proguardFiles(

            getDefaultProguardFile("proguard-android-optimize.txt"),

            "proguard-rules.pro"

        )
```

```
    }

}
compileOptions {
    sourceCompatibility = JavaVersion.VERSION\_1\_8

    targetCompatibility = JavaVersion.VERSION\_1\_8

}
kotlinOptions {
    jvmTarget = "1.8"

}
buildFeatures {
    compose = true

}
composeOptions {
    kotlinCompilerExtensionVersion = "1.4.3"

}
packaging {
    resources {

        excludes += "/META-INF/{AL2.0,LGPL2.1}"

    }

}
}

dependencies {

implementation("androidx.core:core-ktx:1.9.0")
implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
implementation("androidx.activity:activity-compose:1.10.0")
implementation(platform("androidx.compose:compose-bom:2023.03.00"))
implementation("androidx.compose.ui:ui")
implementation("androidx.compose.ui:ui-graphics")
implementation("androidx.compose.ui:ui-tooling-preview")
implementation("androidx.compose.material3:material3")
testImplementation("junit:junit:4.13.2")
androidTestImplementation("androidx.test.ext:junit:1.3.0")
androidTestImplementation("androidx.test.espresso:espresso-core:3.7.0")
androidTestImplementation(platform("androidx.compose:compose-bom:2023.03.00"))
androidTestImplementation("androidx.compose.ui:ui-test-junit4")
debugImplementation("androidx.compose.ui:ui-tooling")
debugImplementation("androidx.compose.ui:ui-test-manifest")
```

}

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

4. Insert the new contents in the following resources and demonstrate their uses in the android application
Android Resources: (Color, Theme, String, Drawable, Dimension, Image)(P2)

1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

package com.example.quest4

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.Image
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.*
import androidx.compose.ui.unit.dp
import androidx.compose.ui.res.*
import androidx.compose.ui.graphics.Color

class MainActivity : ComponentActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    setContent {

        ResourceDemo()

    }

}
```

```
}

@Composable
fun ResourceDemo() {
Column(
    modifier = Modifier

        .fillMaxSize()

        .background(colorResource(id = R.color.backgroundColor))

        .padding(dimensionResource(id = R.dimen.padding\_large)),

    horizontalAlignment = Alignment.CenterHorizontally

){
    Text(

        text = stringResource(id = R.string.app\_title),

        style = MaterialTheme.typography.headlineLarge,

        color = colorResource(id = R.color.textColor)

    )



    Spacer(modifier = Modifier.height(20.dp))



    Image(

        painter = painterResource(id = R.drawable.cat),

        contentDescription = "cat",

        modifier = Modifier.size(200.dp)

    )



    Spacer(modifier = Modifier.height(20.dp))

    }
```

}

2. Modifying strings.xml

Insert the following contents in res/values/strings.xml:

```
<resources>
<string name="app\_name">ResourceDemoApp</string>
<string name="app\_title">Android Resource Demo</string>
</resources>
```

3. Modifying colors.xml

Insert the following contents in res/values/colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>

<color name="backgroundColor">#FFF3E0</color>
<color name="textColor">#000000</color>

</resources>
```

4. Modifying dimens.xml

Create res/values/dimens.xml and insert:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<dimen name="padding\_large">24dp</dimen>
</resources>
```

5. Adding Drawable Resource

Add an image in res/drawable folder.

Example:

sample_image.png

This image will be used in the Image component.

6. Modifying themes.xml

Insert or modify res/values/themes.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
```

```
<style name="Theme.Quest4" parent="Theme.MaterialComponents.DayNight.NoActionBar">

    <item name="colorPrimary">@color/textColor</item>


</resources>
```

7. Configuring build.gradle.kts (Module:app)

```
plugins {
id("com.android.application")
id("org.jetbrains.kotlin.android")
}

android {
namespace = "com.example.quest4"
compileSdk = 35

defaultConfig {
    applicationId = "com.example.quest4"

    minSdk = 24

    targetSdk = 35

    versionCode = 1

    versionName = "1.0"



    testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"

    vectorDrawables {

        useSupportLibrary = true

    }

}

buildTypes {
    release {

        isMinifyEnabled = false

        proguardFiles(
```

```
            getDefaultProguardFile("proguard-android-optimize.txt"),

            "proguard-rules.pro"

        )

    }

}
compileOptions {
    sourceCompatibility = JavaVersion.VERSION\_1\_8

    targetCompatibility = JavaVersion.VERSION\_1\_8

}
kotlinOptions {
    jvmTarget = "1.8"

}
buildFeatures {
    compose = true

}
composeOptions {
    kotlinCompilerExtensionVersion = "1.4.3"

}
packaging {
    resources {

        excludes += "/META-INF/{AL2.0,LGPL2.1}"

    }

}
}

dependencies {

implementation("androidx.core:core-ktx:1.9.0")
implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
implementation("androidx.activity:activity-compose:1.10.0")
implementation(platform("androidx.compose:compose-bom:2023.03.00"))

implementation("androidx.compose.ui:ui")
implementation("androidx.compose.ui:ui-graphics")
implementation("androidx.compose.ui:ui-tooling-preview")
```

```
implementation("androidx.compose.material3:material3")
implementation("androidx.constraintlayout:constraintlayout:2.1.4")
implementation("com.google.android.material:material:1.11.0")
testImplementation("junit:junit:4.13.2")
androidTestImplementation("androidx.test.ext:junit:1.1.5")
androidTestImplementation("androidx.test.espresso:espresso-core:3.5.1")
androidTestImplementation(platform("androidx.compose:compose-bom:2023.03.00"))
androidTestImplementation("androidx.compose.ui:ui-test-junit4")
debugImplementation("androidx.compose.ui:ui-tooling")
debugImplementation("androidx.compose.ui:ui-test-manifest")

}
```

8. Configuring gradle-wrapper.properties

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```
---

5. Create an android application and print “hello world” program(P1)
-->
1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.myapplication

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import androidx.compose.ui.tooling.preview.Preview
import com.example.myapplication.ui.theme.MyApplicationTheme


class MainActivity : ComponentActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    setContent {
```

```
        HelloWorldScreen()

    }

}
}
@Composable
fun HelloWorldScreen() {
Column(
    modifier = Modifier

        .fillMaxSize()

        .padding(16.dp),

    horizontalAlignment = Alignment.CenterHorizontally,

    verticalArrangement = Arrangement.Center

){
    Text(

        text = "Hello World",

         style = MaterialTheme.typography.headlineLarge

    )

}
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
id("com.android.application")
id("org.jetbrains.kotlin.android")
}

android {
namespace = "com.example.myapplication"
compileSdk = 35

defaultConfig {
    applicationId = "com.example.myapplication"
```

```
        minSdk = 24

        targetSdk = 35

        versionCode = 1

        versionName = "1.0"



        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"

        vectorDrawables {

            useSupportLibrary = true

        }

    }

    buildTypes {
        release {

            isMinifyEnabled = false

            proguardFiles(

                getDefaultProguardFile("proguard-android-optimize.txt"),

                "proguard-rules.pro"

            )

        }

    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION\_1\_8

        targetCompatibility = JavaVersion.VERSION\_1\_8

    }
    kotlinOptions {
        jvmTarget = "1.8"

    }
    buildFeatures {
        compose = true
```

```
}
composeOptions {
    kotlinCompilerExtensionVersion = "1.4.3"

}
packaging {
    resources {

        excludes += "/META-INF/{AL2.0,LGPL2.1}"

    }

}
}

dependencies {

implementation("androidx.core:core-ktx:1.9.0")
implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.10.0")
implementation("androidx.activity:activity-compose:1.10.0")
implementation(platform("androidx.compose:compose-bom:2023.03.00"))
implementation("androidx.compose.ui:ui")
implementation("androidx.compose.ui:ui-graphics")
implementation("androidx.compose.ui:ui-tooling-preview")
implementation("androidx.compose.material3:material3")
implementation("androidx.appcompat:appcompat:1.6.1")
testImplementation("junit:junit:4.13.2")
androidTestImplementation("androidx.test.ext:junit:1.3.0")
androidTestImplementation("androidx.test.espresso:espresso-core:3.7.0")
androidTestImplementation(platform("androidx.compose:compose-bom:2023.03.00"))
androidTestImplementation("androidx.compose.ui:ui-test-junit4")
debugImplementation("androidx.compose.ui:ui-tooling")
debugImplementation("androidx.compose.ui:ui-test-manifest")
}
```

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```
---

6. Create a simple Registration Form with the following fields:

- Name (Text Input)
- Email (Text Input)
- Phone Number (Text Input)
- Password (Password Input)
- Confirm Password (Password Input)
- Submit Button

-->

1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.input.PasswordVisualTransformation
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            RegistrationForm()
        }
    }
}

@Composable
fun RegistrationForm(){

    var name by remember { mutableStateOf("") }
    var email by remember { mutableStateOf("") }
    var phone by remember { mutableStateOf("") }
    var password by remember { mutableStateOf("") }
    var confirmPassword by remember { mutableStateOf("") }

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp),
        horizontalAlignment = Alignment.CenterHorizontally
    ){
```

```
Text(
    text = "Registration Form",
    style = MaterialTheme.typography.headlineLarge
)

Spacer(modifier = Modifier.height(20.dp))

OutlinedTextField(
    value = name,
    onValueChange = { name = it },
    label = { Text("Name") }
)

Spacer(modifier = Modifier.height(10.dp))

OutlinedTextField(
    value = email,
    onValueChange = { email = it },
    label = { Text("Email") }
)

Spacer(modifier = Modifier.height(10.dp))

OutlinedTextField(
    value = phone,
    onValueChange = { phone = it },
    label = { Text("Phone Number") }
)

Spacer(modifier = Modifier.height(10.dp))

OutlinedTextField(
    value = password,
    onValueChange = { password = it },
    label = { Text("Password") },
    visualTransformation = PasswordVisualTransformation()
)

Spacer(modifier = Modifier.height(10.dp))

OutlinedTextField(
    value = confirmPassword,
    onValueChange = { confirmPassword = it },
    label = { Text("Confirm Password") },
    visualTransformation = PasswordVisualTransformation()
)
```

```
        Spacer(modifier = Modifier.height(20.dp))

        Button(onClick = { }) {
            Text("Submit")
        }
    }
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android {
    namespace = "com.example.activitydemo"
    compileSdk = 35

    defaultConfig {
        applicationId = "com.example.activitydemo"
        minSdk = 24
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
```

```
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
    implementation("com.google.android.material:material:1.8.0")
}
```

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

7. Create a Registration Form with the following fields:
- Username (Text Input)
- Email Address (Text Input)
- Gender (Radio Buttons with options "Male" and "Female")
- Password (Password Input)
- Submit Button

-->
1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
```

```
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.input.PasswordVisualTransformation
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            RegistrationForm()
        }
    }
}

@Composable
fun RegistrationForm(){

    var username by remember { mutableStateOf("") }
    var email by remember { mutableStateOf("") }
    var password by remember { mutableStateOf("") }
    var gender by remember { mutableStateOf("Male") }

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ){

        Text(
            text = "Registration Form",
            style = MaterialTheme.typography.headlineLarge
        )

        Spacer(modifier = Modifier.height(20.dp))

        OutlinedTextField(
            value = username,
            onValueChange = { username = it },
            label = { Text("Username") }
        )

        Spacer(modifier = Modifier.height(10.dp))

        OutlinedTextField(
            value = email,
            onValueChange = { email = it },
            label = { Text("Email Address") }
```

```
        )

        Spacer(modifier = Modifier.height(20.dp))

        Text("Select Gender")

        Row(
            verticalAlignment = Alignment.CenterVertically
        ){

            RadioButton(
                selected = gender == "Male",
                onClick = { gender = "Male" }
            )
            Text("Male")

            Spacer(modifier = Modifier.width(20.dp))

            RadioButton(
                selected = gender == "Female",
                onClick = { gender = "Female" }
            )
            Text("Female")
        }

        Spacer(modifier = Modifier.height(20.dp))

        OutlinedTextField(
            value = password,
            onValueChange = { password = it },
            label = { Text("Password") },
            visualTransformation = PasswordVisualTransformation()
        )

        Spacer(modifier = Modifier.height(20.dp))

        Button(onClick = { }) {
            Text("Submit")
        }
    }
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

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    id("com.android.application")
    id("org.jetbrains.kotlin.android")
```

```
}

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    compileSdk = 35

    defaultConfig {
        applicationId = "com.example.activitydemo"
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        targetSdk = 35
        versionCode = 1
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            "androidx.test.runner.AndroidJUnitRunner"
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    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
```

```
    implementation("com.google.android.material:material:1.8.0")
}
```

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
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zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```